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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

CS3005-1- MICROPROCESSOR AND EMBEDDED SYSTEMS

(For CS, CB)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which index register in the 8086 is primarily used for operations involving string processing?
A) BP, SP
B) DH, DL
C) BH, BL
D) SI, DI
2. 8086 can access up to?
A) 512KB
B) 2Mb
C) 1Mb
D) 256KB
3. The instruction, MOV AX, 0005H belongs to the address mode
A) register
B) direct
C) immediate
D) register indirect
4. Match the following
a) DB 1) used to direct the assembler to reserve only 10-bytes
b) DT 2) used to direct the assembler to reserve only 4 words
c) DW 3) used to direct the assembler to reserve byte or bytes
d) DQ 4) used to direct the assembler to reserve words
A) a-3, b-2, c-4, d-1
B) a-2, b-3, c-1, d-4
C) a-3, b-1, c-2, d-4
D) a-3, b-1, c-4, d-2
5. Which group of instructions do not affect the flags?
A) Arithmetic operations
B) Logic operations
C) Data transfer operations
D) Branch operations
6. The instruction PUSHF is used to:
A) Push a register onto the stack
B) Push the flag register onto the stack
C) Pop a register from the stack
D) Load a segment register
7. If AX = 1001 0110 1010 1100B and CF = 1, what will be the result of RCL AX, 1?
A) 0010 1101 0101 1001B, CF = 1
B) 0010 1101 0101 1000B, CF = 1
C) 0010 1101 0101 1001B, CF = 0
D) 1001 0110 1010 1101B, CF = 1
8. The instruction RET is used to:
A) Return from subroutine
B) Jump to a memory address
C) Load a segment register
D) Move to the next instruction
9. Which function of INT 16H checks if a key is available?
A) AH = 00H
B) AH = 01H
C) AH = 02H
D) AH = 10H
10. What is the ASCII value of the hexadecimal digit 0x9?
A) 0x39
B) 0x49
C) 0x09
D) 0x1001

11. Set time interrupt from DOS interrupt will store the hour information in which register
 A) CH B) CL
 C) DH D) DL
12. In a program, a Macro is being called 'n' times. Then how many times is the machine code generated for the same?
 A) 1 time B) 'n' times
 C) 'n-1' times D) 'n+1' times
13. Which register in ARM holds the flags indicating the status of the processor?
 A) General-purpose register B) Program Counter (PC)
 C) Stack Pointer (SP) D) Program Status Register (PSR)
14. How does the 7-segment display represent numbers?
 A) Using rows and columns of LEDs B) By turning on specific segments of LEDs
 C) Through alphanumeric LCD cells D) By activating relays
15. Which built-in Arduino function is used for serial communication?
 A) digitalWrite() B) Serial.begin()
 C) analogRead() D) pinMode()
16. What is the primary design philosophy of ARM embedded systems?
 A) Complex Instruction Set Computing (CISC) B) Reduced Instruction Set Computing (RISC)
 C) Very Long Instruction Word (VLIW) D) Superscalar Processing
17. Which pin is used by the 8086 to indicate that an external device is ready to exchange data?
 A) READY B) INTR
 C) ALE D) HLDA
18. Which bit in the MOV instruction encoding specifies whether the data transfer is of a byte or a word?
 A) D (direction) bit B) MOD field bit
 C) W (word) bit D) REG field bit
19. During an interrupt response, which register is the first to be pushed onto the stack by the 8086?
 A) CS (Code Segment) B) IP (Instruction Pointer)
 C) FLAGS D) DS (Data Segment)
20. During a read cycle, when are the address lines present on the multiplexed bus?
 A) After the data is transferred B) During the WR signal activation
 C) When the ALE signal is active D) When the HLDA signal is active

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Identify the addressing modes of the instructions given below and justify the answer with clear explanation. i) MOV WORD PTR[SI], 50H ii) MOV DS:[1000H], 10H iii) MOV AX, Num[BX+DI]	6	L*3	1	2
	b) Write an Assembly level Program to search a key element in a list of 'n' 8-bit numbers using the Binary search algorithm.	6	L3	2	2
	c) Explain the use of instruction queue in 8086 architecture.	4	L2	1	1
2.	a) Describe the roles of various status flags and Control flags in Intel 8086 Microprocessor along with the flag register format. If the two numbers 1100H and 60C9H are added, what are the various flags of Intel 8086 microprocessor set and reset? Justify your answer.	6	L3	1	1

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|-------|--|---|----|---|---|
| b) | How Arithmetic shift instructions work (both left and right)? Differentiate between arithmetic and logical shift instructions. Explain with an example. | 6 | L1 | 2 | 1 |
| c) | Write an Assembly level Program to display your name in front of the message "what is your name". | 4 | L3 | 2 | 2 |
| 3. a) | Explain the working of following 8086 instructions along with example for each:
i) AAM ii) DAA | 6 | L2 | 2 | 1 |
| b) | Explain the following assembler directives with example:
i. ASSUME
ii. EQU
iii. DD | 6 | L2 | 1 | 1 |
| c) | Assume that SI is 3000H and the content of 2FFFH = 04H; content of 3000H = 4EH; Content of 3001H = 9FH. Which value is stored in AX register after execution of the statement MOV AX, [SI]. Justify your answer. | 4 | L3 | 2 | 2 |

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | Briefly explain the Embedded system software components. | 8 | L2 | 4 | 1 |
| b) | Write an Embedded C program to control the speed of rotation of a stepper motor using potentiometer. | 4 | L3 | 4 | 2 |
| c) | Explain the operation of the following DOS INT 21H functions:
i) Function 3AH
ii) Function 2CH | 4 | L2 | 3 | 1 |
| 5. a) | With code snippet, explain Serial.begin(), Serial.available(), Serial.println() functions. | 6 | L2 | 4 | 1 |
| b) | Differentiate between Macro and a Procedure. Write a procedure to read two digit BCD number from keyboard and store in BL register. | 6 | L3 | 4 | 1 |
| c) | Compare and contrast RISC and CISC Architecture. | 4 | L2 | 3 | 1 |
| 6. a) | With a basic layout diagram, explain the ARM program status register by highlighting the functionalities of different fields. | 8 | L2 | 4 | 1 |
| b) | Write an Embedded C program to perform up-down counter on a 7-segment display using switches. | 4 | L3 | 4 | 2 |
| c) | Write an Assembly Language Program to compute the factorial of a positive integer 'n' using a recursive procedure. | 4 | L3 | 3 | 2 |

Unit – III

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|-------|--|----|----|---|---|
| 7. a) | Categorize the following signals into minimum mode signals and maximum mode signals. Write their roles.
(i) S2' to S0' (ii) RQ' / GT1' and RQ' / GT0' (iii) ALE (iv) DEN
(v) HLDA | 8 | L1 | 5 | 1 |
| b) | Explain the coding template for the MOV instruction of 8086 Intel microprocessor and Generate machine code for the following instructions of Intel 8086 microprocessor
MOV AL, [1000] [BX] | 8 | L3 | 5 | 2 |
| 8. a) | List the 8086 signals having common functions in both minimum and maximum mode and explain any four signals in detail. | 10 | L1 | 5 | 1 |
| b) | What is Interrupt Vector and Interrupt Vector Table? Give the various steps involved in interrupt handling. | 6 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
